Related Work

Principia: An Interactive Instrument
for the Three-Body Initial-Condition Manifold

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1 Introduction

Principia is an instrument for exploring the initial-condition (IC) manifold of the planar gravitational three-body problem interactively. It renders a zoomable two-dimensional colour map whose every pixel is one symmetry-reduced IC, integrated forward in time, and whose colour encodes the outcome of that trajectory or a per-trajectory diagnostic. It is a native application first: one Rust engine whose integration kernel is compiled to single precision for the GPU and to double precision for the CPU, with a desktop interface; the first web version compiles that whole native application to run in the browser. The Burrau–Pythagorean research programme uses Principia as its computational platform, for a systematic survey of the infinite family of configurations indexed by primitive Pythagorean triples and of the mass and momentum space around each member of that family.

This document surveys the research traditions on which Principia draws—IC-manifold mapping, shape-space geometry, symbolic dynamics of triple encounters, final-motion classification, chaos diagnostics, statistical theory of outcomes, regularised and symplectic integration, numerical reliability, and GPU computation—and identifies the gaps in the existing literature that motivate the present work.

Several constructs are proposed as part of the present work rather than as established literature: a topological encounter word computed for every initial condition rather than only for periodic orbits (§6); a "projective microscope" that rotates the 2D rendering plane within the 8D manifold (§12); and the use of the Pythagorean-triple family as a structured indexing scheme for three-body IC space (§12). These are labelled as proposed contributions wherever they appear.

A note on the object of study is useful at the outset. The planar three-body problem has a 12-dimensional phase space. Removing the centre of mass (position and velocity) leaves 8; letting the three masses vary on the simplex $\sum m_i = 1$ adds 2; quotienting by rotation and by the scaling symmetry of Newtonian gravity removes 2 more. The space of distinct initial conditions is therefore 8-dimensional: two configuration (shape) coordinates, four momentum coordinates, and two mass-ratio coordinates. Total energy and angular momentum are not fixed: they are functions on this space, derived from each IC rather than imposed as constraints. All prior pixel-scale IC surveys are restricted to a 2-dimensional slice of this space (equal masses, zero or near-zero velocities). Principia is motivated by the need to navigate *families* of such 2D slices inside the full 8-dimensional manifold, making the structure of that manifold—not merely one or two cross-sections of it—the primary object of study.

2 The Three-Body Problem: Fundamentals and History

The three-body problem—determining the motion of three mutually gravitating point masses—was formulated in its modern form by Newton in the *Principia Mathematica* of 1687 [54]. Despite the simplicity of the governing equations, only exceptional families of closed-form solutions are known: Euler's collinear configurations (1763) and Lagrange's equilateral configurations (1772). No general analytic solution is known.

The non-existence of further algebraic first integrals beyond the classical ones was established by Bruns [14], and Poincaré's *Méthodes Nouvelles de la Mécanique Céleste* [56] showed that the problem is generically non-integrable: small perturbations of periodic solutions lead to homoclinic tangles and chaotic behaviour. Fractal basin boundaries in the IC space are what this predicts.

A convergent power-series solution was given by Sundman [66], but it converges too slowly for practical computation. The problem must therefore be treated numerically. Comprehensive introductions from the celestial-mechanics perspective are given by Szebehely [67], Valtonen et al. [73], and Heggie & Hut [31].

Generic bound three-body systems almost always dissolve into a binary and an escaping single body [31]; understanding how the outcome depends on initial conditions, and charting the fractal structure of that dependence across the full IC manifold, is the central problem the work described here addresses.

3 The Pythagorean Three-Body Problem

The configuration now called the *Pythagorean three-body problem* was introduced by Burrau in 1913 [15]: three bodies with masses $m_1 = 3$, $m_2 = 4$, $m_3 = 5$ placed at rest at the vertices of a 3–4–5 right triangle, with each mass equal to the length of the opposite side. Szebehely & Peters [68] gave the first complete numerical solution using regularisation: after a protracted series of close triple approaches, the lightest body escapes, leaving a bound binary of the two heavier bodies. This trajectory has since been a canonical benchmark for three-body codes and a standard demonstration of sensitivity to initial conditions.

Aarseth, Anosova, Orlov & Szebehely [2] investigated the chaoticity structure near the standard Pythagorean IC. They displaced one body at a time to 100 points on each of six circles of radius 5×10^{-8} to 5×10^{-3} about its standard position, recorded the escape angle and the identity of the escaper, and used time-reversed integrations to set aside non-predictable runs. A small neighbourhood of the standard position of each body is a regular region, in which the final outcome varies continuously with the initial position; other regular regions exist, and outside them the outcome changes abruptly. Escape follows a close triple approach, and close triple approaches often made the motion non-predictable even with regularisation—a first map of the IC-space structure near the Pythagorean point, and an early statement of the reliability problem taken up in §10.3. The Burrau problem remains a test case: Boekholt, Moerman & Portegies Zwart [77] solve it to convergence in its Newtonian limit as the control for a relativistic study, and Chitan, Mylläri & Haque [78] integrate all sixteen Pythagorean triangles with hypotenuse below 100, each mass equal to the opposite side, mainly to measure post-Newtonian effects.

The (3,4,5) configuration is the simplest member of an infinite family indexed by primitive Pythagorean triples: every primitive triple (a, b, c) with $a^2 + b^2 = c^2$ yields a distinct right-triangle geometry, and the Burrau mass convention (mass equal to the opposite side, normalised) yields a distinct three-body configuration with rationally related mass ratios. The number-theoretic structure of this family and its implications for the research programme are discussed in Section 12.

4 Mapping the Initial-Condition Manifold

4.1 The Agekyan–Anosova map

The earliest systematic attempt to sample the IC space of the general three-body problem is due to Agekyan & Anosova [4]. For equal masses released (almost) from rest, the configuration is determined, up to scale and rotation, by the shape of the triangle. They fixed two bodies at $(\mp\frac{1}{2}, 0)$ and let the third range over a region D bounded by the positive coordinate axes and the unit circle centred on the first body, $(\xi + \frac{1}{2})^2 + \eta^2 \leq 1$ with $\xi, \eta \geq 0$, which contains every configuration once. Their 100 trajectories, with ICs drawn at random in D, were given small equal and opposite initial velocities rather than none, to avoid a collision at the first close approach; they tabulated each IC with its energy, angular momentum, disintegration time and the perimeter of the closest triple approach. Every completed computation ended in disintegration after a close triple approach. A companion paper [5] continued the study and reprints the initial conditions. This work introduced the paradigm of treating IC space as a two-dimensional region to be surveyed by outcome, and region D underpins all subsequent IC-manifold mapping in this tradition. Li & Liao [43] restate region D at exact rest and use it to search for periodic free-fall orbits.

A key limitation of the Agekyan–Anosova map is that it fixes six of the eight dimensions: equal masses and (near-)zero initial velocities. This surface is a codimension-6 submanifold of the 8-dimensional symmetry-reduced IC space. The dynamics in the remaining six dimensions (independent mass variation and non-zero momenta) are not represented.

4.2 The Lehto et al. pixel map and the Maasai shield

Lehto, Kotiranta, Valtonen, Heinämäki, Mikkola & Chernin [40] carried out a pixel-by-pixel computation over the Agekyan–Anosova domain using approximately 200,000 simulations, colour-coding each IC by the lifetime of the triple system before dissolution. The resulting image resembles a Maasai warrior's shield: broad bands of short-lived systems flanked by intricate swirling structures of longer-lived systems. Zooming in reveals the same band structure repeating at every scale, consistent with fractal basin boundaries; Lehto et al. report a Hausdorff dimension of 1.5 for their map and find 28.6% of positions locally homogeneous at its finest scale, and they map time-reversibility over the same domain.

This study is the closest existing precedent to Principia's rendering task. It is, however, subject to the same restrictions as the Agekyan–Anosova map: equal masses, zero velocities, and a single diagnostic (lifetime), computed in batch with no provision for interactive navigation or zoom. For Principia, the Lehto et al. study both validates the rendering paradigm and defines the narrow 2D subspace within which all prior high-resolution work has been done.

4.3 Converged maps and the structure of the free-fall plane

Boekholt, Portegies Zwart & Valtonen [12] integrated a uniform grid of 1212 initial conditions over the Agekyan–Anosova domain with the arbitrary-precision code Brutus, at accuracies from 10^{-6} down to 10^{-70}, to $t = 10^4$. The same initial condition can live for very different times at the two extremes while the two lifetime maps remain statistically indistinguishable, and the fraction of solutions that cannot be reversed falls as a power law with the numerical accuracy: a map of this domain whose individual pixels have a known accuracy.

Tanikawa and collaborators studied the same plane through its collisions. The initial conditions that lead to a binary collision form curves, and these curves bound the regions of the plane that share a symbol sequence [86, 87, 70]; Tanikawa & Mikkola [88] confirm numerically that triple-collision orbits exist inside the domain, at points where binary-collision

curves meet, and tabulate them. Mikkola & Tanikawa [83] relate the lifetime of each free-fall system to its Lyapunov time, which ties a lifetime map and a Lyapunov map of the same plane together.

Parischewsky, Trani & Leigh [85] map planar free fall near the equilateral (Lagrange) configuration, from the triple-collision singularity out to the hierarchical limit, with outcome maps in the space of side ratios that separate regular from ergodic runs and compare both with the statistical escape theory.

4.4 Beyond the zero-velocity, equal-mass surface

Published maps off this surface are few and narrow. Tanikawa & Mikkola [87] extend their first-symbol maps of the free-fall plane to non-zero angular momentum. Šuvakov & Dmitrašinović [64] search a two-dimensional slice of initial velocities, at fixed collinear positions and equal masses, for periodic orbits. Li, Li & Liao [80] continue one stable periodic-orbit family across a plane of two mass ratios and map its period and stability there—the closest published analogue to a mass chart, though for one orbit family rather than for outcomes. Anosova [76] gave the first outcome statistics for unequal masses in region D, and Valtonen, Mylläri, Orlov & Rubinov [89] integrated 120,000 rotating equal-mass systems from random initial conditions, but these are statistics over random samples, not maps.

4.5 Trani et al. and isles of regularity

Trani, Leigh, Boekholt & Portegies Zwart [72] performed a large-scale IC scan with the *Tsunami* regularised N-body code, running $\mathcal{O}(10^6)$ scattering simulations with unequal masses over a 2D parameterisation that varies the phase of a pre-existing binary and the inclination of the incoming third body. They found that regular (non-chaotic) trajectories occupy a substantial fraction of the scanned IC space, forming connected "isles of regularity" in the chaotic sea, and they separate physical chaos from numerical chaos in doing so. This confirms the picture of a fractal IC-space boundary interspersed with structured regular regions, and demonstrates that modern hardware can sustain IC surveys at million-trajectory scale. Principia's own prototype finds both regimes within a single slice: regions where neighbouring ICs are the same trajectory shifted in orbital phase, and regions where they decorrelate.

The scattering parameterisation of Trani et al. is adapted to hierarchical encounters (one body arriving from far away to interact with a pre-existing binary) and does not cover the non-hierarchical regime in which all three bodies start with comparable separations; its inclination axis also takes the map out of the plane, so only one row of it lies in the planar problem. It is also static: results are computed offline and cannot be explored interactively.

4.6 Fractal basin boundaries

The theoretical framework for fractal basin boundaries was established by McDonald, Grebogi, Ott & Yorke [45], who showed that the boundary between basins of competing attractors is generically fractal when the underlying dynamics are chaotic. Aguirre, Vallejo & Sanjuán [6] showed that in open Hamiltonian systems with three or more escape channels the basin boundaries often satisfy the stronger *Wada* property: every boundary point is a limit point of three or more basins simultaneously. For the three-body problem the relevant escape channels are which body escapes and in which direction; whether its basin boundaries are Wada is a question the IC maps can address directly.

Two tools are standard for quantifying such boundaries. The *uncertainty exponent* (Grebogi, McDonald, Ott & Yorke [25]; see also [45]) measures how the fraction of initial conditions whose outcome flips under an ε-perturbation scales with ε, yielding the uncertainty

dimension of the boundary; *basin entropy* (Daza, Wagemakers, Georgeot, Guéry-Odelin & Sanjuán [21]) measures the Shannon uncertainty of outcomes over a grid and, in its boundary form, gives a test for the Wada property. These are the natural instruments for the IC-space boundaries mapped here, and they carry the quantitative dimension measurements pursued in the companion study.

The KAM theorem (Kolmogorov [36], Arnold [9], Moser [52]) guarantees the persistence of regular islands in nearly-integrable systems. The three-body problem is not nearly integrable, but KAM-like regular islands are nonetheless observed in numerical surveys [72].

5 Shape Space and Symmetry Reduction

5.1 Jacobi coordinates

The standard reduction from Cartesian coordinates uses Jacobi coordinates [35]: the relative vector $\boldsymbol{\rho}$ between an "inner pair" of bodies, and the vector $\boldsymbol{\lambda}$ from the inner-pair barycentre to the third body. These two vectors and their conjugate momenta, together with the decoupled centre-of-mass motion, fully describe the planar three-body system. Principia's decoder builds every IC in mass-weighted Jacobi coordinates: two hyperspherical angles give the configuration at unit moment of inertia, four Jacobi momenta give the velocities, and two logits on the mass simplex give the masses. This chart, the *latent* chart, is Principia's reference coordinate system; every other chart is expressed through it.

5.2 Shape space and the shape sphere

Fixing the overall scale and orientation of the triangle formed by the three bodies leaves a two-dimensional space of triangle *shapes*—the *shape sphere*. Montgomery [50] gives a particularly elegant exposition: the Lagrange equilateral configurations sit at the poles, the collinear configurations on the equator, and the metric is inherited from the mass-weighted kinematic metric on configuration space. The three binary- collision points lie on the equator, interleaved with the three Euler collinear configurations. Choreographic solutions such as the figure-eight [51, 17] trace closed curves on the shape sphere, and their structure is most naturally understood in this representation.

The two hemispheres are mirror images: reflecting every position and momentum maps a system to one on the opposite hemisphere with the opposite angular momentum. Principia's shape-sphere chart covers the whole sphere, every pixel its own system, and treats the mirror as a symmetry of outcomes to be tested rather than an identification to be imposed: for systems released from rest it predicts exactly symmetric outcome maps, and where the chart holds a non-zero angular momentum fixed the two hemispheres are genuinely different systems. The Agekyan–Anosova region is, in this language, a chart on one part of the shape sphere restricted to equal masses.

5.3 Symmetry reduction

Moeckel [48] gives a modern treatment of the natural dynamical reduction of the three-body problem, factoring out translations, rotations and the scaling symmetry of Newtonian gravity. With the masses free on the simplex, the reduced IC space of the planar problem has the 8 dimensions counted in the introduction: 2 shape coordinates, 4 momentum coordinates, and 2 mass ratios. This is the natural domain for a complete survey of three-body IC-manifold structure. All prior IC maps—including the Agekyan–Anosova map and the Lehto et al. pixel map—are restricted to a 2-dimensional equal-mass, (near-)zero-velocity surface within it.

6 Symbolic Dynamics of Triple Encounters

6.1 Classical symbolic dynamics

Symbolic dynamics assigns a finite alphabet to regions of phase space and records the sequence of symbols as a trajectory crosses region boundaries, reducing the continuous dynamics to a discrete combinatorial object. The approach was introduced to celestial mechanics by Alekseev [8], who applied it to the Sitnikov problem (a massless body oscillating through the plane of two equal-mass bodies in circular orbit) and proved the existence of quasirandom trajectories realising any symbol sequence over a two-symbol alphabet.

6.2 Regional coding of the three-body encounter

In the general three-body problem, sequences of visits to regions of the Agekyan–Anosova domain give a coarse coding of trajectories. Heinämäki, Lehto, Valtonen & Chernin [19] used such sequences to compute the Kolmogorov–Sinai entropy across the IC space (see also Section 8.4)—an early example of a symbolic sequence producing a per-IC scalar field over the Agekyan–Anosova map.

6.3 Tanikawa–Mikkola sequences and the homotopy classification

Tanikawa & Mikkola [70, 69] systematically associated symbol sequences with orbits in the free-fall three-body problem, using the equator of the shape sphere—the locus of collinear (syzygy) configurations—as the natural symbol boundary. Their (unsigned) alphabet records the sequence of syzygy crossings, which characterises periodic and collision orbits.

For *periodic* orbits this description has a clean topological form. The reduced configuration space is the shape sphere with the three binary-collision points removed, whose fundamental group is the free group on two generators, F_2. A periodic orbit is a closed loop and carries a free-homotopy class, a *conjugacy class* in F_2. This is the classification of Montgomery [50, 49], applied at scale by Šuvakov & Dmitrašinović [64, 65] and tabulated in the associated orbit gallery [71]; the catalogue tags each family with its reduced word, and these provide exact validation targets for any code that reads such words. An explicit crossing-to-generator dictionary is given by Dmitrašinović, Hudomal, Shibayama & Sugita [22] (their Appendix C, following Tanikawa & Mikkola), who also define the reduced word length N_w used in §12.

Proposed contribution. The present work computes a topological word for *every* initial condition, not only for periodic orbits, and stores it per pixel. Principia records the trajectory's crossings of three branch cuts on the shape sphere, each running from a collision point to a Lagrange pole, with their orientation, and freely reduces the result. Free reduction is legitimate for a generic, non-periodic trajectory because it only removes backtracks, which are contractible detours. The generator and sign conventions are to be transcribed from the literature (Montgomery; Šuvakov–Dmitrašinović), so that where an orbit closes the word reduces to the published F_2 conjugacy class and reproduces the gallery label. Geometrically, the generic word is the homotopy class of the trajectory *arc* relative to its endpoints, an element of the fundamental groupoid of the punctured shape sphere rather than of its fundamental group. The per-pixel word and its evaluation across the non-periodic IC manifold are the contribution here; the periodic-orbit classification and the shape-sphere geometry are those of [70, 69, 50, 49, 64, 65].

Two properties follow from the generic invariant being a path class. First, it depends on both endpoints—the initial shape (the pixel) and the shape at which integration stops—so

generic words are not directly comparable as basepoint-free conjugacy classes are. Second, the final letters depend on the stopping rule (escape criterion or integration horizon); nearby chaotic initial conditions may agree through the bulk of the word and diverge in its tail.

6.4 Heggie's encounter classification

The binary–single scattering problem has been systematically studied by Heggie [30] and Heggie & Hut [31]. They classify outcomes as fly-by, exchange and ionisation. These coarse outcome classes correspond to distinct regions in the IC space of the scattering problem, and the boundaries between them are the three-body analogues of the fractal basin boundaries visible in the Agekyan–Anosova and Lehto maps.

7 Final Motions and Termination

An IC map coloured by outcome is only as good as its definition of an outcome. Chazy [16] classified the possible final motions of the three-body problem as time tends to infinity: hyperbolic, parabolic and elliptic motions of the bodies and their combinations, such as a bound binary with a third body receding hyperbolically or parabolically, and the oscillatory motions, later shown to exist [8]. Principia's outcome classifier is built on this classification, with the caveat that a finite integration can only assign a label *as of the horizon*: "bound so far" is a finite-horizon statement, not a final motion.

The practical question—when an integration may stop because the final motion is settled—is answered in the few-body codes. Fewbody [23] merges bound pairs into a hierarchy of binaries, checks each hierarchy's stability with the Mardling & Aarseth [44] criterion, and stops once every pair of top-level objects is receding, tidal perturbations are below a tolerance, and the top-level system has positive energy. Standish [62] gave sufficient conditions for return, and Agekyan & Anosova [4] already applied an analytic escape criterion (Tevzadze's) to decide when a triple system had disintegrated. Principia's classifier is measured against Fewbody's termination test as its reference. In the prototype, the classifier committed no wrong label on bound ($E < 0$) systems at any playhead and settled earlier than Fewbody's test on the same trajectories—a claim about this implementation, not about Fewbody's purpose, which is termination rather than labelling.

8 Chaos Diagnostics

8.1 Lyapunov exponents and FTLE

The primary quantitative diagnostic of chaos in Hamiltonian systems is the Lyapunov exponent, measuring the exponential rate of divergence of nearby trajectories. Benettin, Galgani, Giorgilli & Strelcyn [10] gave the theoretical foundations and a practical algorithm via periodic reorthonormalisation of tangent vectors evolved by the linearised flow.

For comparing chaos across ICs over a finite horizon, the *finite-time Lyapunov exponent* (FTLE) is more appropriate than the asymptotic exponent. Haller [27] established the FTLE field—computed over a spatial grid of initial conditions—as the standard tool for identifying Lagrangian coherent structures in fluid dynamics, where its ridges delineate the boundaries of qualitatively different flow regimes. Principia computes an FTLE field by the Benettin shadow-trajectory method at its higher quality tiers.

Displaying a chaos indicator as a colour map over a grid of initial conditions is itself a standard technique. Faster-converging variational indicators were developed to separate ordered from chaotic orbits over short horizons: the Fast Lyapunov Indicator of Froeschlé,

Lega & Gonczi [24], MEGNO of Cincotta & Simó [20], and the alignment indices SALI and GALI of Skokos and collaborators [60, 61]. All are routinely drawn as two-dimensional stability maps, and their use in exposing fractal structure is reviewed by Aguirre, Viana & Sanjuán [7]. That literature, however, sits almost entirely within the restricted three-body problem and perturbative stability mapping, where the maps are drawn on a surface of section or over a grid of osculating orbital elements for a test particle in a fixed potential.

We have not found such an indicator field computed over a dense pixel grid of the *general*, non-hierarchical three-body IC manifold, in which all three bodies carry comparable mass and the momentum and mass-ratio degrees of freedom are active. The novelty is that combination—general problem, dense IC-manifold grid, live navigation—not the indicator-as-map technique itself.

8.2 Ensemble spread and the reliability of a pixel

Principia integrates each pixel as a small ensemble of copies at fixed sub-pixel offsets and reports how far the copies' outcomes and final shapes disagree. This spread is both an anti-aliasing signal and a local measure of sensitivity to initial conditions at the scale of the pixel; because the offsets are fixed across pixels and refinement levels, parent and child pixels share their perturbation pattern. It is related to, but not the same as, the uncertainty exponent of §4.6: the uncertainty exponent asks how outcome flips scale with ε, while the spread is read at one ε per zoom level.

8.3 Frequency analysis

Laskar's frequency map analysis [38, 39] identifies regular and chaotic orbits by the drift of fundamental frequencies across sub-intervals of a trajectory: on a KAM torus the frequencies are conserved, in the chaotic region they diffuse. It is itself a mapping technique, routinely drawn over grids of initial conditions in solar-system and galactic dynamics [39]; its relation to Arnold diffusion is discussed by Nekhoroshev [53]. Principia does not currently compute frequency diffusion: an earlier plan to do so was dropped, and it remains a candidate diagnostic.

8.4 Kolmogorov–Sinai entropy in the three-body problem

The Kolmogorov–Sinai entropy—the rate of information creation along a trajectory, equal to the sum of positive Lyapunov exponents—provides a global chaos measure per IC. Heinämäki et al. [19] applied it to the Agekyan–Anosova IC space, producing entropy maps that correlate with the lifetime maps of Lehto et al. [40]. Chernin, Valtonen & Mikkola [18] carried out a related analysis for a restricted four-body system. These entropy maps are early examples of multi-diagnostic rendering of IC-space structure, but are restricted to the equal-mass, zero-velocity subspace (or an analogous restricted configuration).

9 Statistical Theory of Three-Body Outcomes

Because individual three-body trajectories are unpredictable in the chaotic regime, a complementary approach characterises the *distribution* of outcomes over ensembles of ICs. Heggie [30] derived an early statistical theory of binary–single scattering. This was extended by Hut & Bahcall [34] and applied broadly in dense stellar systems.

Stone & Leigh [63] gave a closed-form statistical solution to the non-hierarchical three-body problem under an ergodicity assumption: the predicted outcome distribution agrees well with numerical ensembles for "resonant" encounters (those involving multiple close triple

approaches) but not for the non-resonant ("prompt") fraction, which escapes without full ergodic mixing. Valtonen et al. [89] test the statistical escape theory against rotating systems; Manwadkar, Trani & Leigh [81] characterise the chaotic phase through Lévy flights and the time spent in democratic configurations, and Manwadkar, Kol, Trani & Leigh [82] test the flux-based theory of Kol [79] on escape probabilities for unequal masses. Orlov, Rubinov & Shevchenko [84] measure the algebraic tail of the lifetime distribution of disrupting triples.

Breen et al. [13] trained a deep neural network to predict three-body trajectories at a fraction of the cost of direct integration, demonstrating that the IC space contains learnable structure; accuracy degrades near fractal basin boundaries, exactly the regime where dense diagnostic rendering is most informative. Payot et al. [55] applied active learning to outcome prediction in the Sitnikov problem (§6.1) and found that it fails to improve, and can degrade, predictive accuracy, attributing this to the fractal decision boundary. Together, [13] and [55] delimit the regime where dense per-pixel computation earns its place: an outcome map with learnable structure whose fractal boundaries defeat both interpolation and adaptive sampling.

The tension between the Stone–Leigh statistical theory and the sharp, non-ergodic structure visible in IC maps such as the Maasai shield is an open question. Dense IC maps with outcome and chaos diagnostics side by side are a natural tool for investigating where and how the ergodic approximation breaks down as a function of IC.

10 Numerical Integration and Regularisation

10.1 Symplectic integration

Symplectic integrators, which preserve the symplectic structure of Hamiltonian systems, keep the energy error bounded over long times; Leimkuhler & Reich [41] and Hairer, Lubich & Wanner [26] are the standard treatments. Yoshida [75] showed how to construct symplectic integrators of arbitrarily high order by composing leapfrog steps, giving the fourth- and sixth-order methods Principia offers among its unregularised steppers. At a fixed step size, however, these lose accuracy in close encounters, which are exactly the events that decide three-body outcomes.

10.2 Regularisation

Regularisation removes the collision singularity by a change of coordinates and of the time variable. Kustaanheimo & Stiefel [37] regularised the two-body problem in three dimensions with a four-dimensional spinor transformation. Aarseth & Zare [1] built an eight-dimensional regularisation of the general three-body problem on Kustaanheimo–Stiefel regularisation: collisions of a reference body with each of the other two are regular, and the method stays well behaved while the third distance is not the smallest, which may require occasional transformations back to physical variables to re-label the bodies. Heggie [29] extended their work to a global regularisation, a transformation independent of the configuration under which a collision between any pair is regular; simultaneous collisions of more than one pair and vanishing masses are excepted, and the same construction generalises to the N-body problem. Chain regularisation (Mikkola & Aarseth [46]) and the algorithmic, logarithmic-Hamiltonian regularisation of Mikkola & Tanikawa [47] and Preto & Tremaine [58] extend these ideas to more bodies and to explicit schemes. Aarseth's book [3] covers the methods in practice; modern codes such as REBOUND [59] and Tsunami [72] use variants of them.

Principia regularises from its first version. Its default integrator is Heggie's global regularisation stepped with fourth-order Runge–Kutta; Aarseth–Zare, the logarithmic-Hamiltonian scheme and unregularised steppers are available alongside it, and the step size is limited so

that each close approach is resolved. Aarseth–Zare must re-choose its reference body during a run, and in Principia's prototype that made each pixel's result depend on the times at which the choice is re-examined: changing that cadence moved the outcome map. Heggie's method has no reference body, and with step control held fixed it reduced that sensitivity tenfold; this, rather than a lower energy error alone, is why it is the default.

10.3 Numerical reliability

Li & Liao [42] demonstrated that reliable long-time trajectories in the chaotic three-body problem need extended-precision arithmetic ("clean numerical simulation"). Boekholt & Portegies Zwart [11] showed that, for ensembles of chaotic systems, conventional double-precision integrations can be statistically reliable even when individual trajectories are not, and Boekholt, Portegies Zwart & Valtonen [12] showed that a fraction of three-body systems are irreversible even to perturbations at the Planck length. Hernandez, Hadden & Makino [32] asked the same reliability question of long-term N-body simulations. Principia takes these results as its honesty requirement. Its GPU kernel runs in single precision and its CPU in double precision; the inspector for any clicked pixel runs both, each perturbed by one unit in the last place, and reports the time up to which they agree and the time at which they part, against thresholds taken from this literature. The interface states the predictability horizon this implies, rather than assuming it away.

11 GPU-Accelerated Parallel Computation

Computing an IC map at N^2 pixels, each an independent trajectory integration, is embarrassingly parallel and maps naturally onto GPUs. GPU acceleration for gravitational N-body problems was pioneered by Portegies Zwart and collaborators [57]. The REBOUND code [59] is a widely used CPU-based N-body framework with symplectic integrators.

Principia's integration kernel is written once in Rust and compiled both to the GPU, in single precision, and to the CPU, in double precision. Native GPU access goes through a portable graphics layer, and the browser version reaches the GPU through WebGPU [74]. In the interactive view the field is marched forward in time live, with a coarse image first and detail refined where the field has structure; results quoted from it are confirmed by uniform, full-precision batch runs. What we have not found previously reported is interactive, zoomable IC-manifold visualisation of this kind: a dense per-pixel trajectory integration whose output is a navigable, live map rather than an animation of a single system.

12 The Burrau–Pythagorean Family and Number Theory

The number-theoretic object at the heart of the Burrau–Pythagorean programme is the set of primitive Pythagorean triples, parametrised by Euclid's formula [28]: $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$ for coprime $m > n > 0$ with $m - n$ odd. Each triple defines a distinct Burrau initial condition. As (m, n) range over coprime pairs, the family is a countable, discrete set of points in the 8-dimensional symmetry-reduced IC manifold, and the programme's continuous extension into mass and momentum space is taken separately, as a neighbourhood around each point. The working hypothesis is that fractal structures seen in slices through this family are cross-sections of structures of low codimension in the full manifold, tested by morphing a Burrau slice continuously into unconstrained ones; and the sharpened question is whether the Burrau trajectory lies on the unstable manifold of a nearby periodic orbit.

Proposed contribution: the projective microscope. Principia lets the 2D rendering plane be rotated within the 8D manifold about a locked point. This is a tool for exploration, not a measurement: it buys attention, not truth, and any structure it reveals is confirmed on fixed charts before a claim is made.

The literature on periodic orbits of the three-body problem provides an important reference point. Šuvakov & Dmitrašinović [64] reported 13 new families of planar equal-mass periodic orbits using a topological classification based on the shape sphere. Li & Liao [42] used clean numerical simulation to find more than 600 further families, and in the free-fall problem Li & Liao [43] report 234 collisionless periodic orbits with unequal masses and Hristov, Hristova, Dmitrašinović & Tanikawa [33] revisit the equal-mass collisionless free-fall orbits at scale. These catalogues matter because periodic orbits should appear as distinguished islands of regularity in the IC maps, and because their published initial conditions give exact validation targets.

The most extensive treatment of zero-angular-momentum periodic orbits with their topology is that of Dmitrašinović, Hudomal, Shibayama & Sugita [22], who tabulate more than 200 such orbits, each labelled by its free-group word. They report an approximately linear relation between the scale-invariant period $T_{\text{s.i.}} = T|E|^{3/2}$ and the reduced word length N_w within each of ten sequences that emerge from six linearly stable progenitor orbits; the relation is empirical and approximate. They find 21 linearly stable orbits among those studied, including every progenitor. Linearly stable orbits are natural candidates for the islands of regularity expected in the IC maps, and the tabulated orbits give an end-to-end validation set: a correct integrator and word reader should reproduce the published words and sequences.

The family itself has been integrated before: Chitan, Mylläri & Haque [78] run all sixteen Pythagorean triangles with hypotenuse below 100, one trajectory each in the Newtonian case, as a test set for relativistic corrections. We have not found prior dynamical-systems work that uses the Pythagorean-triple family as an indexing scheme for three-body configurations, or that maps the IC space around its members.

13 Validation Targets

Several published solutions serve as exact or reference tests for an instrument of this kind: the Lagrange and Euler analytic solutions; the figure-eight [51, 17]; the Burrau problem as solved by Szebehely & Peters [68]; the periodic-orbit catalogues above [64, 65, 71, 43, 33]; the triple-collision points tabulated inside region D [88]; the lifetime grid of [12] at accuracy 10^{-70}; and the Agekyan–Anosova initial conditions [4, 5], whose tabulated angular momenta follow directly from their stated ICs. The tabulated energies in [4] do not follow from the paper's own energy formula; that discrepancy is recorded rather than resolved.

14 Summary of Gaps

The literature reviewed above establishes a clear lineage from Agekyan & Anosova (1967) through Lehto et al. (2008) and Trani et al. (2024): the IC manifold of the three-body problem can be rendered as a colour-coded 2D image, revealing fractal basin boundaries, regular islands, and entropy structure. The theoretical foundations in shape space, symbolic dynamics, final motions, Lyapunov theory, and statistical mechanics are well developed. The following gaps are identified.

Coverage of the IC manifold. Almost all prior pixel-scale IC surveys of the general problem lie on one 2-dimensional surface of the 8-dimensional symmetry-reduced IC manifold:

equal masses at (near-)zero velocity. The exceptions are few and narrow: first-symbol maps at non-zero angular momentum, a velocity slice searched for periodic orbits, a mass plane mapped for one orbit family, and scattering maps that fix a hierarchical geometry. We have not found outcome or chaos maps that let the momentum and mass dimensions vary in general, or that move between such slices.

Interactivity and zoom. All prior IC maps were computed offline in batch. We have not identified a tool that permits interactive navigation—zoom, pan, chart switching, a live time axis—within an IC map that is computed as it is viewed.

Simultaneous diagnostics with stated reliability. Prior maps render a single diagnostic per computation (usually lifetime or escape identity), and do not report per pixel whether that diagnostic can be trusted. We have not found prior work that computes outcome, FTLE, a topological encounter word and an ensemble spread together over a dense IC grid, with each pixel's precision-limited reliability stated.

Symbolic encounter invariants. Existing symbolic dynamics uses coarse regional codings, unsigned shape-sphere sequences (Tanikawa–Mikkola), or—for periodic orbits only—the free-homotopy classification in F_2 (Montgomery; Šuvakov & Dmitrašinović). A topological encounter word computed for *every* initial condition, reducing to the periodic-orbit class where the orbit closes, has not, to our knowledge, been reported as a dense per-pixel field.

Variable-mass IC maps. Mass ratios appear as a map axis only for the continuation of a periodic-orbit family, and unequal masses otherwise only as fixed parameters or in outcome statistics. The effect of mass variation on basin topology, fractal dimension, and regular-island structure is not systematically documented in the literature we have reviewed.

Structured family surveys. The Pythagorean family has been integrated member by member, one trajectory each. We are not aware of a systematic survey of how the IC structure around each member varies across the family, or across any other number-theoretically indexed family of configurations.

Scope of the review. This review is necessarily incomplete; the gaps identified are those apparent from the literature accessible to us at the time of writing. Unpublished code, conference proceedings, or adjacent-field work may address one or more of these points, and any strong gap claim should be treated as a research hypothesis rather than an established fact. Even with Principia, interactive 2D slice navigation is not exhaustive coverage of the 8-dimensional IC manifold: the tool makes selected 2D cross-sections navigable, and the structure of the full manifold can only be inferred from families of such cross-sections.

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